# Customer Transaction Analysis

*Exploratory Data Analysis Report*

## Executive Summary

This report analyses a banking customer file (5,000 accounts) and a transaction file (219,843 transactions over one month) to understand how customers use their accounts and where the bank can grow revenue. Two thirds of accounts (65%) are dormant, fewer than one in five customers use any digital channel, and activity is dominated by transfers and cash withdrawals. Active accounts carry almost all of the transaction value, so moving dormant customers back to active use is the clearest revenue opportunity. Only 17% of customer accounts could be linked to the transaction file, so findings that combine the two files are directional rather than conclusive.

## Objective

To understand the bank's customer base and transaction behaviour, and to identify practical ways to reactivate dormant accounts, increase transaction activity and grow revenue.

## Data Description

| Dataset | Size | What it contains |
|---|---|---|
| customers_data | 5,000 rows (4,467 unique customers, 4,995 unique accounts) | Account status, product, customer type, balance, digital channel enrolment, card status |
| transactions_data | 219,843 rows (9,164 accounts, 8,783 customers) | Amount, debit or credit flag, timestamp, transaction description |

The transactions cover 31 August 2020 to 1 October 2020, which is about one month. Amounts are in the bank's local currency (the data does not state which). The analysis was done in Python using pandas, matplotlib and seaborn.

## Data Cleaning and Preprocessing

- **File formats:** the customer file is tab-separated and the transaction file is comma-separated, so each was loaded with the correct delimiter.
- **Dates:** opening date, last transaction date and transaction timestamps were converted to proper date formats. The timestamps came in mixed formats and were parsed individually.
- **Columns dropped:** email (68.3% missing) was too sparse to use, and region contained a single value for every row, so it carried no information.
- **Missing values:** blanks in card status (62.5%), restriction code (29.4%), address (13.7%) and product group (1.4%) were labelled "Not Specified" instead of being deleted or guessed, because a blank often means "not applicable", such as no card issued.
- **Balance outliers:** working balances range from about -1.97 billion to +102 million, which is not realistic for most accounts. These values were flagged (739 accounts, 14.8%) but

not changed, and the median balance (about 958) was used instead of the average. In total, 363 accounts (7.3%) have a zero or negative balance.
- **Linking the files:** the two files were joined on account ID and customer ID. Only 868 accounts (17.4% of customer accounts and 9.5% of transaction accounts) appear in both files, so 21,379 transactions (9.7%) could be linked to a customer profile.

## Key Findings

### Customer base

| Account status | Accounts | Share |
|---|---|---|
| Dormant | 3,260 | 65.2% |
| Active | 1,456 | 29.1% |
| Inactive | 284 | 5.7% |

- Individual customers make up 82.1% of accounts (4,106), followed by corporate (6.6%), salary (3.7%), minor (2.9%) and enterprise (2.0%) customers.
- The product mix is heavily savings-based: savings accounts are 75.4% of accounts (3,769) and current accounts 17.2% (858). Domiciliary accounts and retail loans are small, with 141 domiciliary current, 53 domiciliary savings and only 61 retail loan accounts.

### Digital channel adoption

| Channel | Share of accounts enrolled |
|---|---|
| USSD | 17.9% |
| OnePay | 16.9% |
| OneBank | 9.4% |

### Transaction behaviour

- Debits make up 71.2% of transactions (156,511) and credits 28.8% (63,279).
- Credits are larger but less frequent: the median credit is about 16,968 compared with about 7,070 for a median debit, roughly 2.4 times larger.
- Transfers dominate activity, followed by cash withdrawals and card purchases.

| Transaction type | Count | Share |
|---|---|---|
| Transfer (TRF) | 87,004 | 39.6% |
| Transfer in | 54,069 | 24.6% |
| Cash withdrawal | 32,128 | 14.6% |
| Point of sale purchase | 23,072 | 10.5% |
| Bill payment | 12,682 | 5.8% |

| Transaction type | Count | Share |
|---|---|---|
| Cash deposit | 3,717 | 1.7% |

### Activity by account status (linked accounts only)

| Account status | Linked accounts | Median transactions per account |
|---|---|---|
| Active | 742 | 15 |
| Inactive | 59 | 5 |
| Dormant | 67 | 3 |

Active accounts are 85.5% of the linked accounts but generate 91.0% of the linked transaction value, and they transact five times as often as dormant accounts.

## Insights

1. **Dormancy is the biggest revenue opportunity.** Roughly two out of three accounts (65.2%) are dormant. These customers already have an account and completed onboarding, so reactivating them costs far less than acquiring new customers.
2. **Activity is where the money is.** Active accounts transact about five times as often as dormant ones and account for 91% of linked transaction value, so account status is a real behavioural signal the bank can target.
3. **Digital channels are barely used.** OneBank is used by only 9.4% of accounts, and OnePay and USSD by 16.9% and 17.9%. Low digital engagement sits alongside high dormancy, so digital enrolment is a practical, controllable lever.
4. **Cash is still heavily used.** Cash withdrawals (14.6% of transactions) are more common than card purchases (10.5%), which means higher cash handling cost for the bank and a missed opportunity to grow card and bill payment revenue.
5. **The product mix is narrow.** Three quarters of accounts are savings accounts and only 61 accounts are retail loans, so there is clear room to cross-sell loans, domiciliary accounts and other products.
6. **Inflows are larger than outflows.** Customers receive fewer but larger credits and spend in many smaller debits, which suggests balances are available that the bank could put to work through savings or investment products.

## Recommendations

1. **Run a targeted dormant account reactivation campaign.** Start with dormant accounts that transacted most recently or still have a digital channel enrolled, using SMS, call-backs and in-branch prompts, with a small incentive such as a bonus for the first transactions after reactivation.
2. **Push digital enrolment together with reactivation.** Make OneBank, OnePay and USSD sign-up part of every reactivation and onboarding contact, because digital access gives customers a reason to use the account regularly.
3. **Shift customers from cash to cards and bill payments.** Offer cashback or rewards on point of sale and bill payment transactions to reduce cash handling cost and increase fee income.

4. **Cross-sell to active customers.** Offer retail loans, fixed deposits and domiciliary accounts to active savings customers, and loan products to the 183 salary customers.
5. **Put larger inflows to work.** Offer automatic savings or fixed deposit options to customers who receive large credits, so idle balances earn income for the bank and interest for the customer.
6. **Improve the customer and transaction data link.** Only 17% of customer accounts match the transaction file, so a complete link would let future campaigns be targeted using the whole customer base.

## Limitations

- The transactions cover only about one month, so seasonal patterns cannot be assessed.
- Findings that combine both files are based on the 868 linked accounts and are directional.
- Balance values include extreme figures that should be confirmed with the data owner before being used in any financial reporting.

## Conclusion

The bank's customer base is mostly dormant, with low digital adoption and a savings-heavy product mix, while transaction activity is concentrated in a small group of active accounts. The biggest opportunities are to reactivate dormant customers, drive digital and card usage, and cross-sell additional products to engaged customers. Acting on these recommendations would increase transaction volumes and revenue without the cost of acquiring new customers. Fixing the link between the customer and transaction data would make future targeting more accurate.